Augmented City VPS

Michela di Vietro (CTO)



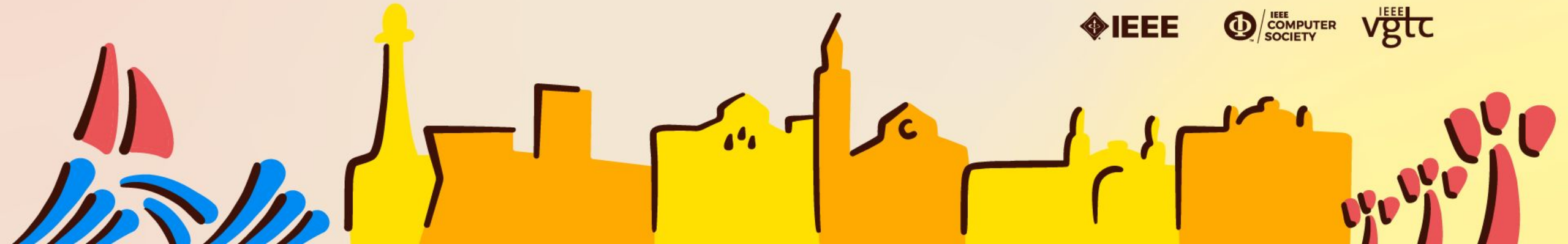
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ISMAR 2026

October 5 - 9, Bari, Italy

The 25th International Symposium on Mixed and Augmented Reality

IEEE IEEE COMPUTER SOCIETY IEEE VGTC



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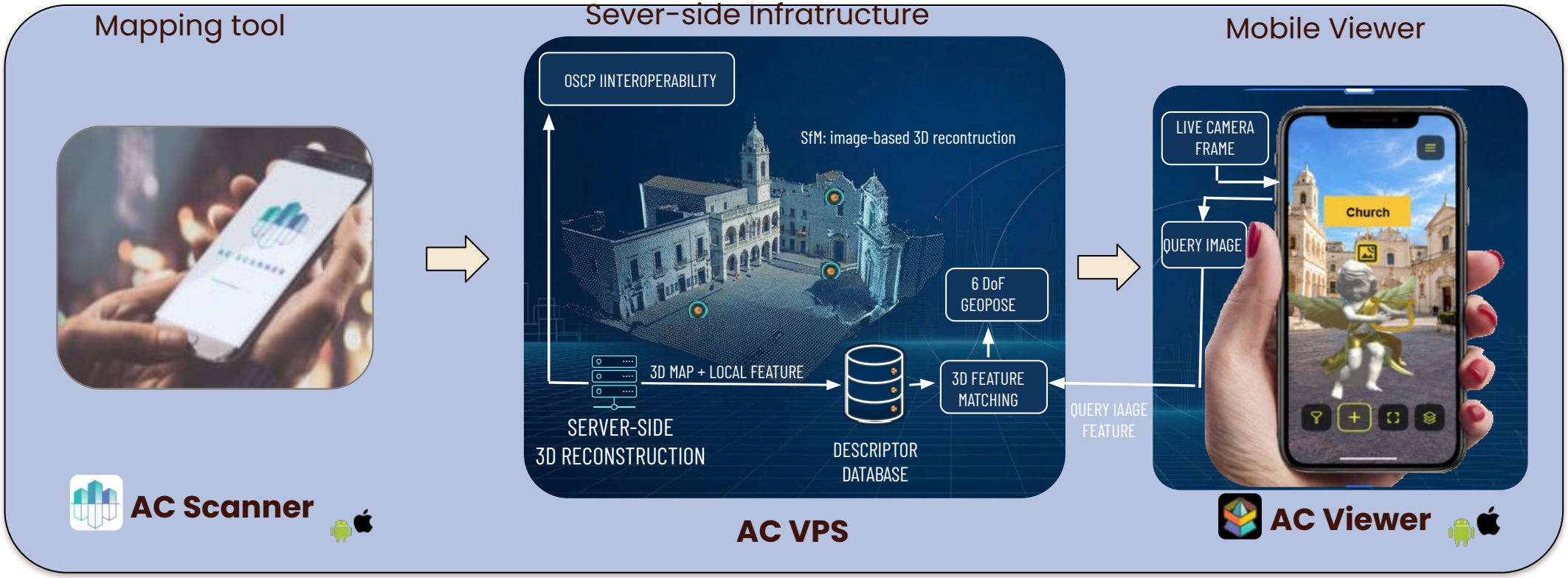
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AC Platform

A suite of apps designed to work with GEOPose-based Augmented Reality

3 apps to support the workflow:

- ***AC Scanner*** - mobile mapping and data capture for outdoor (any size of location from a small site to a city scale) and indoor
- ***Management System*** - Map-based georeferencing and digital-twin/content management
- ***AC Viewer*** - mobile AR visualization and interaction



Augmented City VPS

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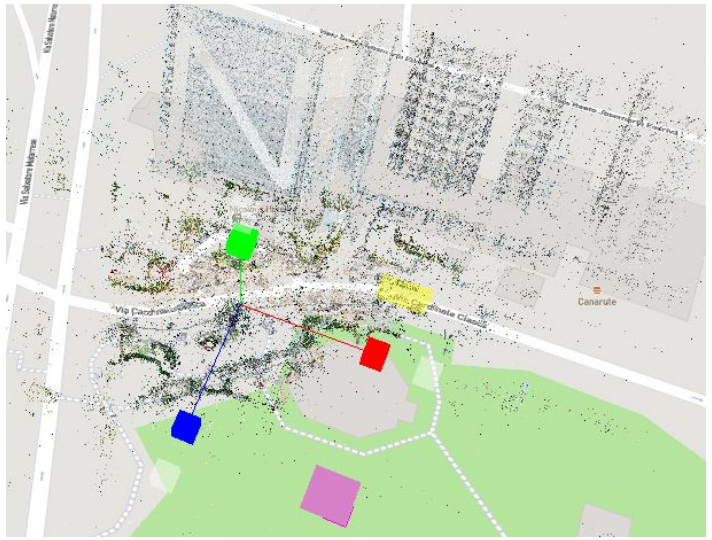
Key Components:

- **Smartphone Mapping** — Creation of 3D point clouds from *standard smartphones* using SfM algorithms; *no LiDAR or specialized hardware required*.
- **Visual Localization** — Fast localization using ***proprietary computer-vision algorithms*** to locate the query image within the 3D Point Cloud and estimate the camera pose.
- **Spatial Content Placement** — Accurate placement and real-time interaction with virtual content *on-site across different devices or remotely*.

Augmented City VPS

Key Technical Features

- Mapping via **any smartphone**, processing and storage on cloud server
- No size limit for Point Clouds
- Full processing on the server, **no need to download point clouds to the device**



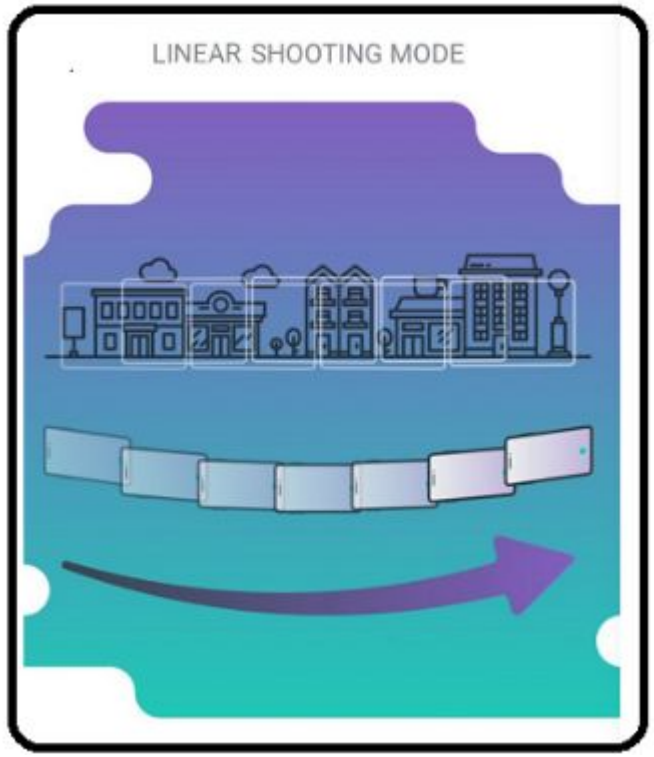
- **Automatic Point Cloud Merging**
- Works in all weather conditions and brightness changes, offers a localization accuracy of 10-30 cm
- Highly robust solution, supports up to **75% of scene change**
- Ultra-fast localization (less than 0.1 sec on the server)
- **Support for OGC GeoPose 2.0 protocol**
- Rating system and educational toolbox



AC Scanner - Mapping Procedure

How to create a 3D map (Digital twin) of a real-world location

LINEAR scanning





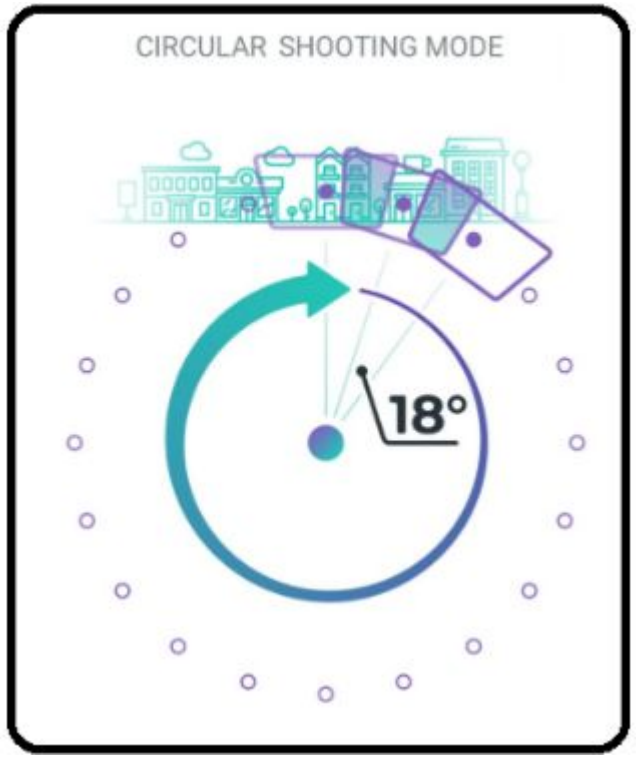


CALIBRATION passage automatically create a *Point Cloud with real-world dimensions*.

AC Scanner - Mapping Procedure

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CIRCULAR scanning

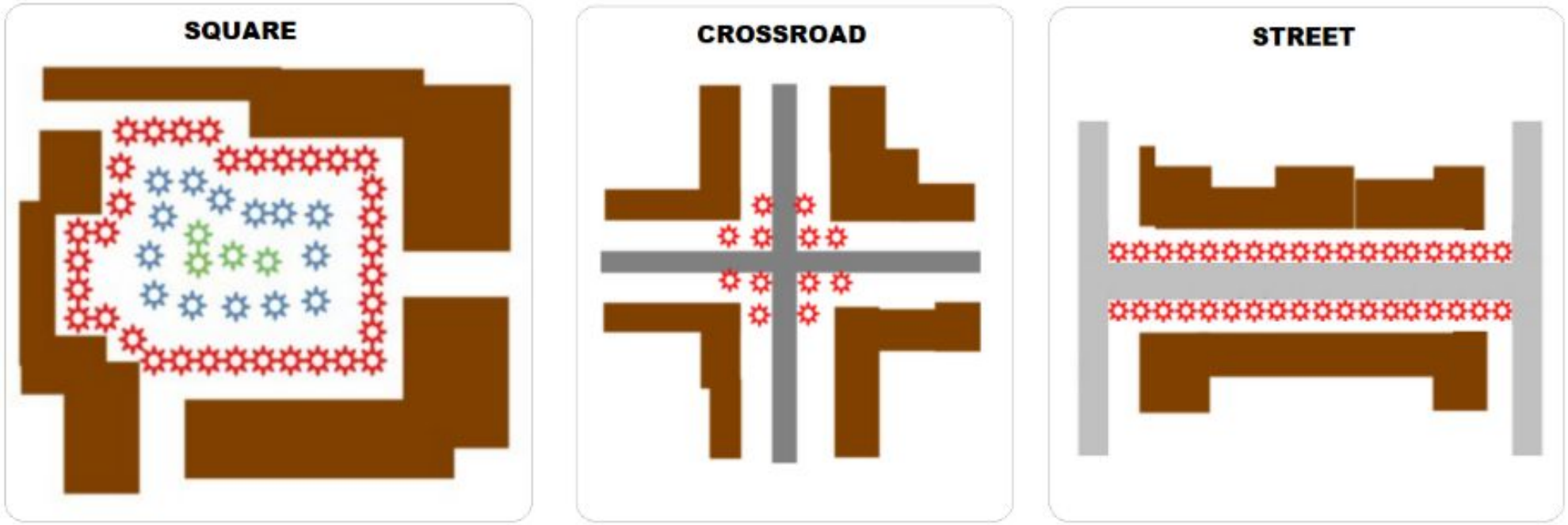




Scan points at the same distance from the reference facade.

AC Scanner - Mapping Procedure

Mapping Approaches for Different Area Types

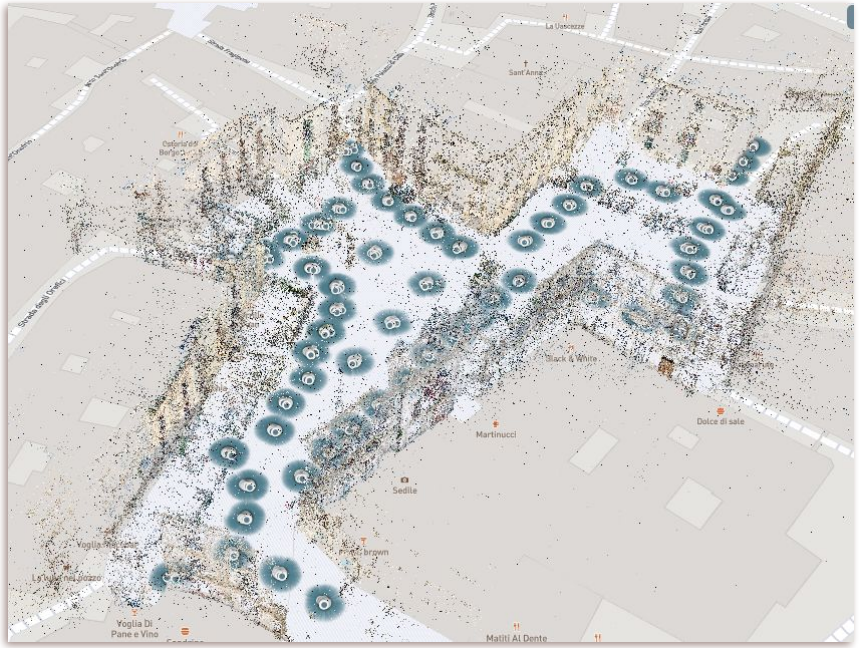


- point at 5 meters from the facade
- point at 10-15 meters from the facade
- point at more then 15 meters from the facade

AC Scanner – Mapping Procedure

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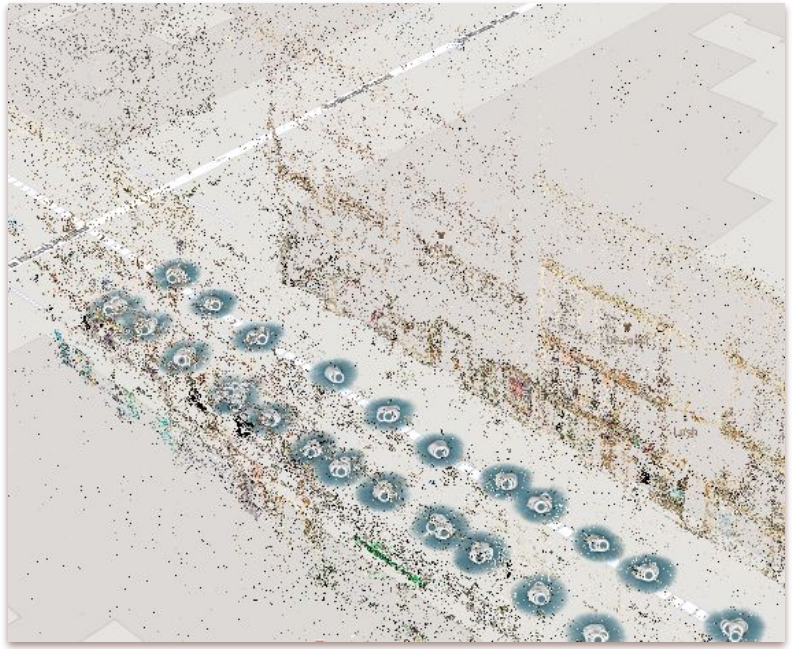
Mapping result for different Area Types: **3D Point Cloud & camera pose**



square



cross



street

AC Scanner – Spatial Mapping

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Some Examples of our 3D Point Clouds – in 230 cities in the world

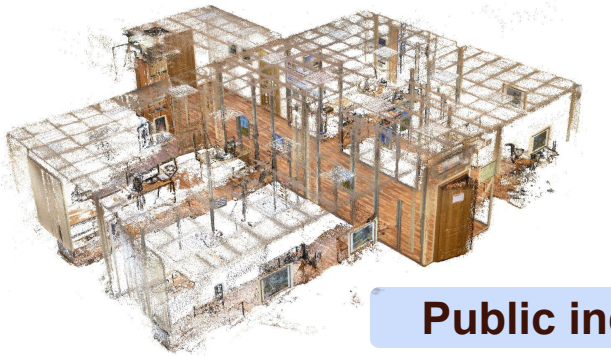
Outdoor and indoor scanned locations



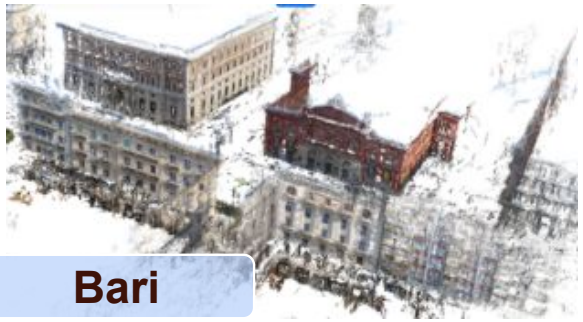
Pisa



Acireale



Public indoor space



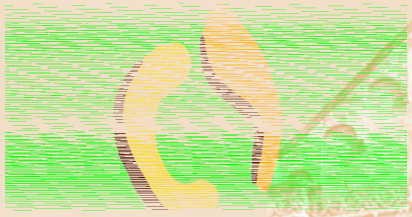
Bari



Sanremo



A Picture Gallery indoor space



THANKS FOR THE ATTENTION

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